

Solution Architecture

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Team ID	NM2025TMID03796
Project Name	LEASE MANAGEMENT

1. Architecture Overview (high-level)

- Primary platform: Salesforce (Lightning Experience) — houses data model, UI, automation, reports.
- Optional middleware: Heroku / MuleSoft / AWS Lambda or any iPaaS for complex integrations.
- External services: Payment gateway, e-signature provider, SMS/WhatsApp gateway, accounting system.
- CI/CD: SFDX + Git + CI pipeline (GitHub Actions / Azure DevOps / CircleCI).
- Monitoring & audit: Salesforce Shield (Event Monitoring, Field Audit) or native Audit Trail + external logging.

2. Logical Components

1. Presentation Layer

- Lightning Record Pages, Lightning Components (LWC) for Property, Tenant, Lease, Payment, Maintenance, Feedback.
- Mobile support via Salesforce Mobile App / responsive LWCs.

2. Application Layer (Business Logic)

- Declarative: Flows, Process Builder (prefer Flows), Validation Rules, Approval Processes.
- Programmatic: Apex controllers, scheduled Apex (for periodic tasks), Apex REST for custom APIs.

- Trigger patterns: Use Apex triggers only when necessary; wrap logic in handler classes.

3. Data Layer

- Custom Objects: Property__c, Tenant__c, Lease_Agreement__c, Payment__c, Maintenance__c, Feedback__c.
- Relationships: Lookups / Master-Detail as designed (1:N).
- Indexing & selective queries: Add custom indexes for large orgs (External Ids).

4. Integration Layer

- Outbound: Callouts to payment gateway, e-sign provider, SMS gateway.
- Inbound: Webhooks to create Payment records, notify lease events.
- Middleware: Use Heroku/MuleSoft for orchestrating multi-step flows, buffering retries, and transforming payloads.
- API: Expose Apex REST endpoints for portals or external apps.

5. Storage & Files

- Attachments & documents: Files (Salesforce Files) + Content Version.
- Large documents (if needed): Store in external S3-compatible service and link URL in Salesforce.

6. Reporting & Analytics

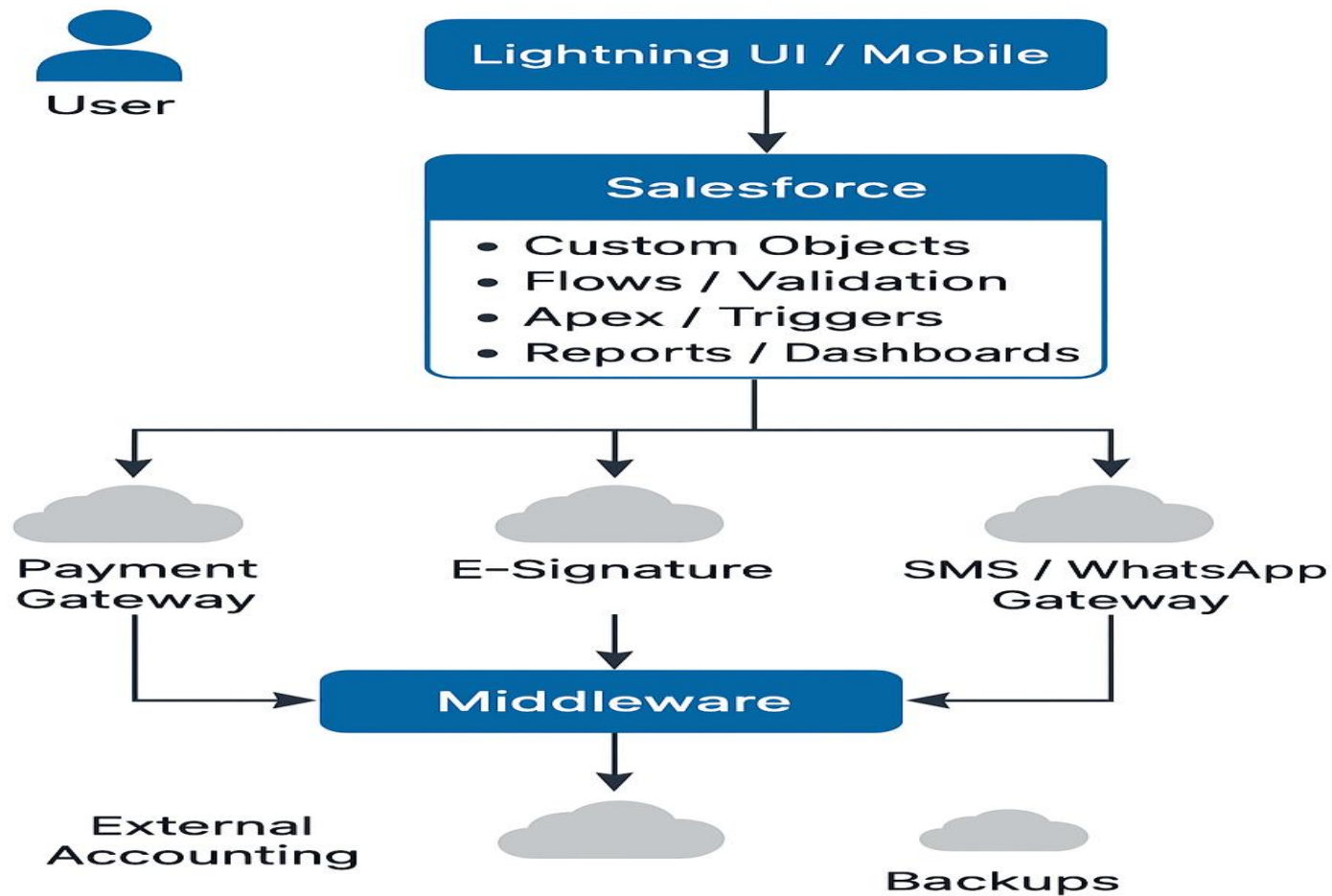
- Salesforce Reports & Dashboards (operational views).
- Einstein / Tableau CRM for advanced analytics (optional).

7. Security & Compliance

- Authentication: Salesforce SSO / OAuth for external users; multi-factor authentication.
- Authorization: Profiles, Permission Sets, Sharing Rules, Role Hierarchy.
- Field-level security & record-level sharing for sensitive fields (rent, bank details).

8. CI / Deployment

- SFDX source-driven development (metadata or unlocked packages).
- Git branches for feature/dev/test; CI pipeline for validation and deployment to orgs.



3. Typical Data / Event Flow (sequence)

1. Admin creates Property__c.
2. Tenant record created via UI or portal (Tenant__c).
3. Admin creates Lease_Agreement__c linking Property & Tenant.
4. Flow calculates schedule; scheduled Apex creates Payment Due reminders.
5. Reminder triggers email/SMS (via standard Email + external SMS gateway).
6. Tenant pays via Payment Gateway → gateway webhook → middleware → Apex REST → creates Payment__c.
7. On full payment, Flow updates Lease status to Active; generates receipt (Salesforce PDF) and stores in Files.
8. Maintenance requests create Maintenance__c and initiate approval if cost > threshold.
9. At lease end, Flow triggers renewal notification and archives records as needed.

4. Integrations (recommended patterns)

- Payment Gateway: Use secure tokenized API; implement asynchronous webhooks and a middleware layer for retries.
- E-signature: DocuSign or Adobe Sign connector (prebuilt Salesforce packages available).
- SMS / WhatsApp: Use Twilio or Gupshup via middleware (avoid storing phone credentials).
- Accounting: Push monthly invoices and payments to external accounting via middleware (batch or near-real-time).
- Portal / Mobile: Use Experience Cloud for tenant/owner portal or expose REST APIs for custom mobile apps.

5. Security & Data Governance

- Configure profiles and permission sets:
 - Admin: full CRUD

- Owner: view their property & leases
 - Tenant: limited view of own lease & payments
- Use sharing rules to limit record visibility by role/territory.
- Enable MFA and enforce strong password policies.
- Use Field-Level Encryption for sensitive financial info; avoid storing full card numbers.
- Enable Login IP ranges and session timeouts for Admin profiles.
- Regular backups: Weekly full export + daily incremental backups to external storage.

7. Non-Functional Requirements & Recommendations

- Scalability: Design with bulk-safe Apex, selective SOQL, indexed fields. For orgs with >100k records, use skinny tables / external objects as needed.
- Performance: Cache read-heavy data (Custom Settings / Custom Metadata); avoid synchronous callouts on UI transactions.
- Availability: Salesforce SLA for uptime; have a runbook for outages and manual override processes.
- Recoverability: Scheduled backups + sandbox tests for restore procedures.
- Maintainability: Modular Apex, use Platform Events for decoupling, document Flows, use unlocked packages.

8. Monitoring & Observability

- Enable Debug logs and Apex exception emails for errors.
- Use Salesforce Shield / Event Monitoring for tracking access and suspicious activity.
- Build dashboards for system health: failed flows, failed integrations, queued jobs, long-running transactions.
- Configure alerts (email/Slack) from middleware/CI pipeline for deployment issues or integration failures.

9. Deployment & Delivery Plan (summary)

- Phase 1: Data model + basic UI + create/read/update/delete flows (MVP).
 - Phase 2: Payment integration + notifications + reporting.
 - Phase 3: Portal, e-sign, accounting sync, analytics.
 - Use SFDX + Git; deploy first to a sandbox, then to staging, then production.
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10. Implementation Checklist (quick)

- Create custom objects & fields with correct relationships
- Add validation rules (dates, amounts)
- Build Flows: Lease creation, reminders, renewals
- Implement Apex for complex logic & APIs
- Integrate payment gateway (webhook + middleware)
- Configure Experience Cloud (optional) for tenant portal
- Setup reports & dashboards
- Setup permission sets & sharing rules
- Enable backups and monitoring
- Setup CI/CD